Ethernet Standards and Specifications

Assignment Answers

Q1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards



The RJ-45 connector has 8 pins, numbered 1 to 8 from left to right when the contacts face up and the latch clip faces down. Each pin connects to one of the 8 wires of a twisted-pair cable. T568A and T568B are the two wiring standards that decide which colour goes to which pin.

Difference between T568A and T568B

- The blue pair (pins 4 and 5) and the brown pair (pins 7 and 8) are the same in both standards.
- Only the orange and green pairs are swapped. In T568B, pins 1 and 2 are orange and pins 3 and 6 are green. In T568A, pins 1 and 2 are green and pins 3 and 6 are orange.
- T568B is the most commonly used standard for ordinary patch cables. The important rule is to use the same standard on both ends of a normal cable.

Q2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Specification	**10BASE-T**	**100BASE-TX**	**1000BASE-T**
Common name	Ethernet	Fast Ethernet	Gigabit Ethernet
IEEE standard	802.3i	802.3u	802.3ab
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	UTP twisted pair, Cat3 or higher	UTP twisted pair, Cat5 or higher	UTP twisted pair, Cat5e or Cat6 (Cat5 also works in many cases)
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45	RJ-45	RJ-45
Wire pairs used	2 pairs (pins 1, 2, 3, 6)	2 pairs (pins 1, 2, 3, 6)	All 4 pairs (all 8 pins)

Q3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Think about how transmit and receive pairs are wired in each cable type.

Cable to use: A crossover cable.

Reason: On a laptop's Ethernet port, pins 1 and 2 are used to transmit (TX) data and pins 3 and 6 are used to receive (RX) data. When two laptops are joined directly, the transmit pins of one laptop must connect to the receive pins of the other laptop.

- Straight-through cable: pin 1 goes to pin 1, pin 2 to pin 2, pin 3 to pin 3 and pin 6 to pin 6. The transmit pins of laptop 1 would connect to the transmit pins of laptop 2, so both laptops would be talking and nobody listening, and the link would not work (this type is meant for connecting a computer to a switch or router, where the switch ports are wired the opposite way).
- Crossover cable: the pairs are swapped, so pins 1 and 2 on one end go to pins 3 and 6 on the other end. Now the transmit pair of one laptop reaches the receive pair of the other, and communication works.

Pin on Laptop 1 end	Goes to pin on Laptop 2 end	Purpose
1 (TX+)	3 (RX+)	Laptop 1 transmit to Laptop 2 receive
2 (TX-)	6 (RX-)	Laptop 1 transmit to Laptop 2 receive
3 (RX+)	1 (TX+)	Laptop 2 transmit to Laptop 1 receive
6 (RX-)	2 (TX-)	Laptop 2 transmit to Laptop 1 receive

How to make it: A crossover cable is made by using T568A on one end and T568B on the other end. For Gigabit speed, the pairs on pins 4, 5, 7 and 8 are also crossed (4 to 7 and 5 to 8).

Note: Most modern laptops have Auto-MDIX, which automatically changes the transmit and receive pins, so even a straight-through cable usually works. However, according to the basic rule, a crossover cable is the correct one for connecting two similar devices.

Q4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	Where it is found	Likely Ethernet standard
WiFi router (LAN ports)	Home or college	1000BASE-T. Most modern routers have Gigabit LAN ports. Older or budget routers may have only 100BASE-TX.
Desktop PC / laptop	College computer lab or home	1000BASE-T. Modern PCs and laptops have Gigabit Ethernet network cards.
Smart TV	Home living room	100BASE-TX. Many smart TVs have a 100 Mbps Ethernet port, which is enough for video streaming.